

Enzo Harquin

PhD Student in Computer Science

📍 Champs-sur-Marne, France 🌐 eharquin

Summary

- PhD student at the Gaspard-Monge Computer Science Laboratory (LIGM), specializing in geometric algebra, computer vision, and computer graphics.

Experience

Gaspard-Monge Computer Science Laboratory (LIGM), PhD Student Gustave Eiffel University, France
Research on geometric algebra and its applications to computer vision and computer graphics, with a particular focus on Projective Geometric Algebra.

Gaspard-Monge Computer Science Laboratory (LIGM), Research Intern Champs-sur-Marne, France
Parametric curves and surfaces in geometric algebras. Apr 2024 – May 2024
• Supervised by Vincent Nozick and Venceslas Biri. 2 months

Education

Gustave Eiffel University, Image Science Champs-sur-Marne, France
• Graduated with honors. Sept 2023 – 2024
• Ranked joint first among 19 students.

University of Technology of Compiègne (UTC), Computer Engineering, Software Engineering specialization Compiègne, France
Sept 2020 – Sept 2023

Projects

Voxel Rendering Engine in Vulkan/C++ Nov 2023 – Mar 2024
Research project for the Master's Degree in Image Science.
• Literature review, implementation, and comparative study of voxel rendering methods.
• Rasterization, ray tracing, and ray marching.

MultiVector
Interactive visual environment for geometric algebra constructions, research, exploration, and teaching.
• Expression-based construction and interactive manipulation of geometric objects.
• Support for multiple algebra models, including VGA, PGA, CGA, ACGA, and CCGA in two and three dimensions.
• Export of constructions as SVG and PNG images.

Skills

Research Areas

Programming Languages

Scientific Tools and Environments

Languages

French
Native speaker

English

C1 level — Cambridge English Linguaskill certification